Fikile Mkhize 0210210785080

0781087624

14 July 2026

REPRESENT ANALYSE AND CALCULATE SHAPE AND MOTION IN 2 AND 4L3 DIMENSIONAL SPACE IN DIFFERENT CONTEXTS

US9016

Formative assessment

Question 1

a) Small-hole gauge→ Telescopic gauge / Internal micrometer / Small hole gauge

b) Eectrical Values→ Multimeter

c) Chemical Analysis→ Burette / Pipette / Measuring cylinder

d) Short Distances in Meters →Vernier calipers / Micrometer / Steel rule

e)Amount of milk in a container→ Measuring cylinder / Graduated jug

Question 2

a) Name of instrument : Vernier Caliper

b) 1. Outside jaws - measure outside diameter

2. Inside jaws - measure inside diameter

3. Depth rod/probe - measure depth

4. Main scale - shows whole mm/cm

5. Vernier scale - shows decimal part

6. Lock screw - locks the reading

7. Sliding jaw

8. Thumb wheel/fine adjuster

c) Si unit:meter (m),Usually measured in mm

d)Place the object between the jaws. Read the value on the main scale first, then find which line on the vernier scale lines up with the main scale. Add both readings to get the exact measurement.

Fikile Mkhize 0210210785080

Question 3

a)Square and Rectangle both have 4 angles of 90°

b) 3-D shape with 4 triangular faces + 1 square face → square-based Pyramid

c) Shapes with diagonals that always cross at right angles: square,Rhombus,kite

Question 4

a) Cube side = 2cm

Surface Area = 6×2^2

$= 6 \times 2^2 = 6 \times 4$

$= 24cm^2$

Volume $= sides^3$

$= 2^3$

$= 8cm^3$

b) Enlarged by scale factor 5. New side $= 2 \times 5 = 10cm$

New Surface Area = 6×10

$= 600cm^2$

New Volume = 10^3

$= 1000cm^3$

Question 5

Rectangle A: 6cm x 2cm Rectangle B: 24cm x 8cm

a) Area of A $= 6 \times 2 = 12cm^2$

b) Area of B: $24 \times 8 = 192cm^2$

c)Scale factor= 24/6=4 or 8/4=4

d) Times bigger: Scale factor$^2 = 4^2 = 16$ times bigger

Fikile Mkhize 0210210785080

Question 7

Wall = 24m, Base = 7

Ladder^2=24^2+7^2

x^2=576+49

x^2=625

x=/sqrt(625)

x=25m

Length of ladder = 25m